

SHELBY L LEVEL

Status: Research Analyst

Field: Public Health, Data Analytics, Data Visualization

Techs: R, Git, Shiny, Quarto, RMarkdown, GitHub, GitLab, Docker

Email: slevel20@gmail.com

Website: <https://www.shelbylevel.org>

GitHub: <https://www.github.com/shelbylevel>

Summary

Health data analytics, visualization, and dashboard enthusiast looking for an opportunity to continue my learning journey and expand my technical skill set with R.

Skills

Technical

- * R Shiny Dashboard Development and Maintenance: Experience building and maintaining multi-tab modular dashboards
- * Data visualization: Expertise in creating and improving static and interactive data visualizations and maps using a variety of R packages (ggplot2, plotly, highcharter, leaflet, sf)
- * Sharing Data: Skilled at literate computing and creating HTML documents, PDFs, Word documents, websites, and presentations using R Markdown and Quarto (revealjs, beamer, Bootswatch, bookdown)
- * Experience with creating a Quarto website and hosting via Netlify and GitHub pages (GH-pages)
- * Functional HTML, CSS, SCSS, SASS, and LaTeX coding experience
- * Experience in securing, cleaning, and preparing data for analysis, storage, or presentation in R (tidyr, dplyr)

Communication & Leadership

- * Experience assisting with the onboarding of new dashboard staff across projects
- * Experience working in a collaborative development environment using version control (Git, GitHub, GitLab, GitKraken)
- * Excellent oral and written communication skills
- * Project management experience including client and stakeholder relationship building, maintenance, and deliverables

Education

Biostatistics Graduate Certificate - The Ohio State University

Jan. 2025 - Current

- * Ongoing coursework

Master of Public Administration - Ohio University

Jul. 2023 - May 2024

- * Graduate Research Assistant

Bachelor of Science in Public Health - University of Toledo

Aug. 2020 - May 2023

- * Magna cum laude

Work Experience

Senior Research Analyst (S3) - Ohio Colleges of Medicine Government Resource Center

Feb. 2026 - Present

- * Serving as a technical lead for the development and maintenance of interactive dashboards, analytical applications, and reporting systems supporting statewide public health and healthcare initiatives
- * Directing the design of scalable, reproducible analytics workflows and data products using R, Shiny, Quarto, Git, and related technologies
- * Lead accessibility initiatives across organizational data products
- * Contributing to survey analytics activities including sample design support, weighting methodologies, measure development, statistical analysis, and dissemination of findings
- * Mentoring analysts and developers through onboarding, technical guidance, code review, and documentation improvements
- * Driving the adoption of consistent branding, design systems, and user-centered development practices to improve trust, usability, and stakeholder engagement with data products

- * Developed, maintained, and improved interactive dashboards using R Shiny
- * Worked to improve the digital accessibility of public-facing, internal, and client-facing project dashboards to comply with WCAG guidelines
- * Assisted with onboarding new staff across dashboard teams
- * Improved documentation usability and navigation with dynamic and interactive options such as Quarto, Revealjs, Mark-down, and other HTML formats
- * Collaborated with the communications team to expand brand identity using cohesive styling and formatting across project deliverables and documents
- * Learned new statistical methodologies by assisting with autocorrelation testing, measure construction, survey weighting, ad hoc requests, etc.
- * Led sample frame creation for survey projects and assisted with analysis and visualization of survey results

Research Assistant - Ohio University, Voinovich School of Leadership & Public Service

Jul. 2023 - May 2024

- * Assisted with gathering 2012-2021 Ohio Medicaid Assessment Survey (OMAS) data for the Appalachian addendum and the Disability addendum of the 2023-2025 State Health Assessment (SHA)
- * Created a brief report for the Regional QI Hub Quality Improvement Project (QIP) - Diabetes Institute, Ohio University on the connections between social determinants of health and chronic health outcomes and behaviors at the county and census tract levels in Ohio
- * Analyzing survey data, social vulnerability indices, other Census Bureau data, and mapped transportation options and ADA compliant resources in Athens county for the Creating Healthy Communities Ohio Hub (Athens City County Health Department)
- * Building interactive dashboards and applications with Shiny

Research Staff - Ohio University, Ohio Alliance for Population Health (OAPH)

Jul. 2023 - May 2024

- * Analyzing CDC WONDER mortality data to identify health disparities in Ohio across geographic regions, time, and demographics
- * Authoring several reports (diabetes, firearm mortality, disease mortality, etc.) focused on Ohio most generally and Appalachian Ohio in particular
- * Providing ad hoc analytics and visualization support for OAPH members and clients

Health Policy Intern - Health Policy Institute of Ohio (HPIO)

Jan. 2023 - Jun. 2023

- * Used secondary data to create data visualizations by Ohio's counties, zip codes, and census tract levels
- * Provided ad hoc analytics and visualization support for HPIO members and clients
- * Reviewed data entry, statistical calculations, and proper citation formatting for reports and other materials as part of quality control measures

Data Analytics Intern - Ohio University, Ohio Alliance for Population Health (OAPH)

Aug. 2022 - Dec. 2022

- * Used R to gather and clean Census Bureau data, National Survey on Drug Use and Health (NSDUH) data, Ohio Department of Health (ODH) data (various), and CDC WONDER mortality data
- * Profiled drug overdoses at the national, state, and county levels (Lucas County, Ohio in particular) with series of illustrations including tables, charts, maps

Student Research Assistant - University of Toledo, Department of Medicine

May 2022 - Aug. 2022

- * Replicated HIV-1 Protease molecular dynamics simulations in vitro
- * Induced mutations in the HIV-1 Protease, purified each mutated protease including four mutations and one wild type, and ran small angle x-ray scattering (BioSAXS) to confirm protein confirmation in simulated data
- * Learned wet lab techniques in recombinant DNA transformation and protein purification
- * Presented project updates to primary investigator and research team through weekly team meetings

Data Projects

Social Determinants of Health in Ohio's Counties and Census Tracts - Ohio University

Dec. 2023 - Apr. 2024

- * This project involved comparing and exploring commonly used Social Determinants of Health (SDOH) indices and their association with health outcomes in Ohio's counties and census tracts

* Assisted with gathering demographic, Health Professional Shortage Areas (HPSAs), Medically Underserved Areas (MUAs), physician workforce, health status, and health outcomes data for Maryland and the Baltimore City and County metro area

* Used SAS to clean and analyze data from Our World in Data and the Worldbank International Database

* Performed regression analyses to determine if COVID-19 vaccination rates significantly reduced new COVID-19 cases or excess mortality in approximately 185 nations during Fall of 2021

Certifications

* National Commission for Health Education Credentialing Credential ID: 38648

Leadership & Community Engagement

* Nominated to become a member of the 2022-2023 KLAR Leadership Academy through the University of Toledo Neff College of Business and Innovation

* Strengthened leadership, teamwork, communication, and critical thinking skills by participating in learning and service events on campus and within the local community